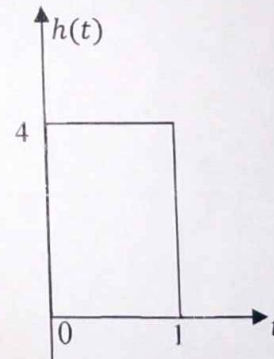
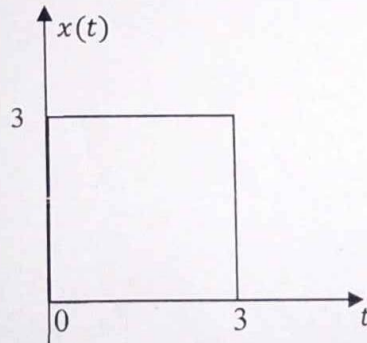


→ (a) A system response is given as: $y(n) = nx^2(n)$. Two students from ICT engaged in a heated debate about a system's nature. The first student argued it's an LTI system, while the other insisted it's system is an LTV system. As an EEE student enrolled in EEE 206, Signals and Systems arbitrate between them and:

- (i) Clarify who is correct and provide the reasoning **(3marks)**
- (ii) State one property for Time-Invariant systems **(1mark)**
- (b) Determine the stability of system b (i)) and provide your stability assessment **(2marks)**
- * (i) $y(t) = 3x^5(t)$
- (ii) Compute the amplitude for system b (i) **(2marks)**
- (iii) Illustrate the final response of the system b (i) in a graphical representation **(2marks)**

Question Two

(a) Perform the convolution operation on the given pair of signals the two signals $x(t)$ and $h(t)$



- (a) Provide a visual representation of the ultimate convoluted signals. (2marks)
(b) Determine the area encompassed by the resulting signals. (1mark)
(c) Identify and describe a practical context where convolution finds application. (1mark)

$3x^5 - 0$
 $x^5 - 0$
 $x^5 - 0$
 $x^5 - 0$



Instruction: Answer only Three (3) Questions Time: 120 minutes

Question One

(a) A student from the biomedical department at FUTA seeks clarification on two systems, causal and non-causal, from you as an ICT student. As part of the explanations, determine the causality and non-causality of the two systems in (a) and (b):

(i) $y(n) = 4x(t) + 5x(t - 3)$ (4marks)

(ii) $y(n) = 4x(t) + 5x(t - 5) + 6x(t + 6)$ (4marks)

(iii) Justify your output in (a) and (b) with reasons (2marks)

(iv) Give one property and one area of application for each system (2marks)

(v) Which out of the two systems you would recommend for future operations and why? (2marks)

(b) Determine the stability of system b (i) and provide your stability assessment:

(i) $\int x dt$ (3marks)

(ii) Compute the amplitude for system b (i) (1mark)

(iii) Represent the final response of the system b (ii) in pictorial form (2marks)

Question Two

The head of the department, ICT, gave an assignment to students in the biomedical department. One of them, in an effort to finding solutions to this assignment needs your assistance as 200-level ICT student who has taken ICT 204, signals and systems and scored A in the course:

The system response is given as $y(t) = [U(t + 1) - U(t - 3)] * [U(t + 1) - U(t - 1)]$

(i) Tell the student the convolution property you want to deploy in solving this problem and why? (2marks)

(ii) Determine the convolution of the two unit steps signals given in (a) (8marks)

(iii) Compute the area covered by the resultant signal (3marks)

(iv) Produce the final convoluted signal in pictorial form (4marks)

(v) What is the length of period, T, of the final resultant signal in (a)? (3marks)

Question Three

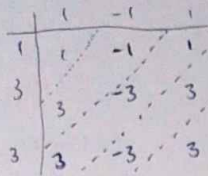
(a) In the given signal: $x(t) = 4 \cos(4t \pm 30) + 8 \sin(16t \pm 60)$

(i) Compute the amplitude of the fundamental components of the given signal (4marks)

(ii) Find the frequency of the fourth harmonics (3marks)

(iii) Calculate the amplitude of the third harmonics (3marks)

(b) A signal $x(n) = [1, -1, 1]$ is convolved with an unknown signal $h(n)$, the convolution result is noted to be $y(n) = [1, 2, 1, -2, 3]$ assuming a linear convolution is deployed.

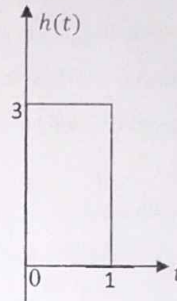
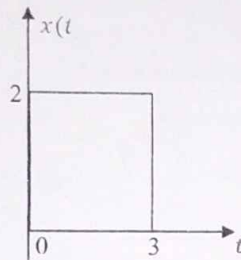


Determine:

- (i) the number of samples in $h(n)$ (2marks)
- (ii) actual samples in $h(n)$ (6marks)
- (iii) $h(2)$ in $h(n)$ and state the method employed to determine the actual samples in $h(n)$ (2marks)

Question Four

- (a) Perform the convolution operation on the given pair of the two signals $x(t)$ and $h(t)$ (10marks)



- (b) Provide a visual representation of the ultimate convoluted signals. (5marks)
- (c) Determine the area encompassed by the resulting signals. (3mark)
- (d) Identify and describe a practical context where convolution finds application. (2mark)

Question Five

(a) Two students, X and Y from MNE department, have some ambiguities regarding the periodicity of some signals. As an EEE student who enrolled for EEE 206 signals and systems, kindly clear their ambiguities by following these steps:

- (i) Explain what it is meant by a signal is periodic and aperiodic signal, giving an example of each and one area of application. (4mark)

(b) Determine whether or not each of the following signals is periodic and give a reason for your claim. If a signal is periodic, determine its fundamental period:

(i) $x(t) = \cos \frac{\pi}{3} t + \sin \frac{\pi}{4} t$ (4mark)

(ii) $x(t) = \cos t + \sin \sqrt{2} t$ (4mark)

(c) Explain the following:

- (i) Why Fourier transform (2mark)

- (ii) Find the Fourier transform of the given signal $x(t) = e^{-a|t|}$ $a > 0$ (6marks)